Example 1 — Acute Decompensated HFrEF With Pneumonia and Acute Kidney Injury

Patient: 72-year-old man
Admission diagnosis: Acute hypoxic respiratory failure due to acute on chronic systolic heart failure and community-acquired pneumonia
Discharge diagnoses:

- Acute on chronic HFrEF, EF 30%
- Community-acquired right lower lobe pneumonia
- Acute kidney injury, improved
- Hypertension
- Type 2 diabetes mellitus
- Chronic atrial fibrillation
- Chronic anemia

Hospital Course

The patient presented with 5 days of progressive dyspnea, orthopnea, bilateral lower-extremity edema, and productive cough. On arrival, oxygen saturation was 84% on room air. Chest radiograph demonstrated pulmonary vascular congestion with a right lower lobe infiltrate. BNP was elevated, and echocardiogram demonstrated LVEF approximately 30%, decreased from 40% on prior study.

He was treated with IV loop diuretics with improvement in respiratory status and peripheral edema. He initially required supplemental oxygen but was weaned to room air prior to discharge. He was also treated for presumed community-acquired pneumonia with ceftriaxone and azithromycin, with clinical improvement. Blood cultures remained negative. Sputum culture grew normal respiratory flora.

Creatinine increased from 1.2 to 1.6 mg/dL during diuresis and subsequently improved to 1.3 mg/dL prior to discharge. Guideline-directed heart failure therapy was adjusted during admission. ACE inhibitor was held during the acute kidney injury and transitioned to a lower-dose ARB at discharge. Beta blocker was continued. Mineralocorticoid receptor antagonist was deferred pending repeat renal function and potassium testing.

The patient was clinically euvolemic at discharge and able to ambulate without significant dyspnea.

Discharge Medications

New medications

- **Furosemide 40 mg by mouth once daily** for volume management.
- **Losartan 25 mg by mouth once daily** for heart failure and blood pressure control.
- **Empagliflozin 10 mg by mouth once daily** for heart failure/diabetes management.
- **Amoxicillin-clavulanate 875/125 mg by mouth twice daily for 3 additional days** to complete treatment for pneumonia.

Continue

- Metoprolol succinate 50 mg once daily.
- Apixaban 5 mg twice daily.
- Atorvastatin 40 mg nightly.
- Metformin 500 mg twice daily, provided renal function remains stable.

Stopped/held

- Lisinopril — discontinued because therapy was changed to losartan.
- Spironolactone — held pending repeat renal function and potassium.
- Ibuprofen/other NSAIDs — avoid because of heart failure and recent kidney injury.

Discharge Instructions

- Follow a **low-sodium diet**, generally targeting approximately 2 grams of sodium daily.
- Monitor daily weight at approximately the same time each morning.
- Keep a written record of weight and symptoms.
- Follow fluid recommendations provided by the heart-failure team.
- Complete the entire antibiotic course even if feeling better.
- Avoid NSAIDs unless specifically approved by the treating clinician.
- Obtain a basic metabolic panel before or at the early outpatient follow-up to reassess creatinine and potassium.
- Follow up with primary care and cardiology within approximately 1 week.

Return Precautions

Seek urgent medical attention for:

- New or worsening shortness of breath, especially at rest.
- Chest pain or pressure.
- Fainting or severe dizziness.
- Rapidly worsening leg or abdominal swelling.
- Significant decrease in urine output.
- New confusion or severe weakness.
- Fever or worsening cough despite antibiotics.

The patient should contact the heart-failure team promptly for a **rapid increase in weight or worsening swelling/breathlessness**, according to the individualized weight-management plan provided at discharge.

Nursing Discharge Teaching / What Nursing Typically Relays

Nursing would typically reinforce:

1. **Medication reconciliation:** review which medications are new, continued, stopped, and temporarily held.
2. **Diuretic teaching:** explain that furosemide increases urination and discuss timing it earlier in the day when possible.
3. **Daily weights:** demonstrate how and when to weigh and what action to take if weight rises according to the physician's prescribed threshold.
4. **Heart-failure symptoms:** reinforce monitoring for increasing edema, orthopnea, shortness of breath, or rapid weight gain.
5. **Antibiotic adherence:** emphasize completing the antibiotic course.
6. **Bleeding precautions:** because the patient takes apixaban, review signs of significant bleeding and the importance of notifying clinicians about new medications.
7. Confirm that the patient understands **follow-up appointments and laboratory testing**.
8. Use teach-back: ask the patient to explain the medication changes and when they would seek medical attention.

Example 2 — Sepsis From Complicated Urinary Tract Infection With Obstructing Stone

Patient: 64-year-old woman
Admission diagnosis: Sepsis secondary to complicated UTI/pyelonephritis
Discharge diagnoses:

- Sepsis due to complicated urinary tract infection
- Left ureteral calculus with hydronephrosis
- Status post ureteral stent placement
- Acute kidney injury, resolved
- Type 2 diabetes mellitus
- Hypertension

Hospital Course

The patient presented with fever, rigors, left flank pain, nausea, and dysuria. Initial laboratory testing demonstrated leukocytosis, elevated lactate, and acute kidney injury. CT abdomen/pelvis demonstrated a left ureteral stone with associated hydronephrosis.

She received IV fluids and empiric broad-spectrum antibiotics. Given concern for infected obstruction, urology performed urgent left ureteral stent placement for decompression. Urine culture subsequently grew a susceptible strain of *E. coli*. Blood cultures remained negative. Antibiotics were narrowed based on culture and susceptibility results.

Renal function returned near baseline following decompression and treatment of infection. The patient remained afebrile and hemodynamically stable for >24 hours before discharge.

The ureteral stent remains in place and requires outpatient urologic management for definitive stone treatment and eventual stent removal.

Discharge Medications

New medications

- **Cefpodoxime 200 mg by mouth twice daily for 7 additional days** to complete antibiotic therapy.
- **Tamsulosin 0.4 mg by mouth once daily** while ureteral stent remains in place, unless otherwise directed by urology.
- **Acetaminophen as needed** for mild pain or fever, within the maximum daily dose instructed by the treating team.

Continue

- Amlodipine 5 mg daily.
- Atorvastatin 20 mg nightly.
- Home diabetes medications as reviewed on the final medication reconciliation.

Stopped/held

- Lisinopril — temporarily held pending outpatient renal function and blood-pressure reassessment.
- NSAIDs — avoid because of recent kidney injury unless specifically approved.

Discharge Instructions

- Complete all prescribed antibiotics.
- Maintain adequate oral hydration unless another clinician has specifically restricted fluids.
- Follow up with urology for definitive stone management and ureteral-stent removal/exchange.
- Do not assume that absence of pain means the stent can remain indefinitely; it requires planned follow-up.
- Have kidney function rechecked as directed.
- Monitor blood glucose according to the established diabetes plan.

Return Precautions

Return immediately for:

- Fever or shaking chills.
- Worsening flank or abdominal pain.
- Persistent vomiting or inability to keep medications down.
- Confusion, fainting, severe weakness, or difficulty breathing.
- Markedly decreased urine output.
- Symptoms suggesting recurrent urinary obstruction or infection.

Nursing Discharge Teaching / What Nursing Typically Relays

Nursing would typically emphasize:

- **Finish the antibiotic exactly as prescribed.**
- Explain common urinary symptoms associated with a ureteral stent, while distinguishing these from warning signs of infection.

- Confirm that the patient knows the **urology follow-up date or scheduling process**.
- Reinforce hydration instructions.
- Review medication changes and specifically identify medications that were stopped or held.
- Teach the patient to monitor for fever and worsening urinary/flank symptoms.
- Review glucose monitoring if relevant to the patient's diabetes plan.
- Confirm that prescriptions were obtained or that the patient knows where/how they will be filled.
- Use teach-back to confirm understanding of antibiotic duration and return precautions.

Example 3 — Acute Decompensated HFpEF With Atrial Fibrillation and Cellulitis

Patient: 81-year-old woman
Admission diagnosis: Acute on chronic HFpEF with volume overload and right lower-extremity cellulitis
Discharge diagnoses:

- Acute on chronic diastolic heart failure
- Atrial fibrillation with rapid ventricular response, resolved
- Right lower-extremity cellulitis
- Hypertension
- Chronic kidney disease stage 3
- Type 2 diabetes
- Chronic lower-extremity edema

Hospital Course

The patient presented with worsening bilateral lower-extremity edema, approximately 10-pound weight gain, orthopnea, and erythema/warmth of the right lower leg. She was found to be in atrial fibrillation with rapid ventricular response.

She received IV diuresis with substantial improvement in edema and dyspnea. Rate control was achieved after adjustment of her beta-blocker regimen. Echocardiogram showed preserved LVEF with findings consistent with HFpEF.

The right lower-extremity cellulitis was treated initially with IV antibiotics, followed by oral therapy after clinical improvement. There was no evidence of abscess or osteomyelitis. Renal function remained stable near baseline.

At discharge, she was breathing comfortably on room air, ambulatory with her usual assistive device, and had improving lower-extremity erythema.

Discharge Medications

New medications

- **Torsemide 20 mg by mouth once daily**, replacing prior intermittent diuretic use.
- **Metoprolol succinate 75 mg once daily** for rate control.
- **Cephalexin 500 mg four times daily for 4 additional days** to complete cellulitis treatment.
- **Empagliflozin 10 mg once daily** for heart failure management.

Continue

- Apixaban 5 mg twice daily, with dose to be taken according to the final medication reconciliation.
- Amlodipine as instructed.
- Atorvastatin nightly.
- Home diabetes regimen as reconciled.

Stopped

- Furosemide — replaced by torsemide.
- Previous short-acting metoprolol regimen — replaced by extended-release formulation.

Discharge Instructions

- Weigh yourself daily.
- Follow a low-sodium diet.
- Elevate the affected leg when resting as instructed.
- Complete the cephalexin course.
- Monitor the cellulitis for decreasing redness, warmth, and swelling.
- Take anticoagulant exactly as prescribed and do not discontinue it without medical advice.
- Follow up with cardiology and primary care.

- Obtain renal function/electrolytes as ordered after the diuretic change.

Return Precautions

Seek urgent evaluation for:

- New chest pain.
- Severe or rapidly worsening shortness of breath.
- Syncope or near-syncope.
- Sustained rapid or irregular heartbeat associated with symptoms.
- Rapidly worsening edema or weight gain.
- Cellulitis spreading beyond the marked area or developing rapidly increasing pain.
- New fever, chills, confusion, or severe weakness.
- Significant bleeding while taking anticoagulation.

Nursing Discharge Teaching / What Nursing Typically Relays

Nursing commonly reviews:

- The **old diuretic versus the new diuretic**, making clear that the patient should not take both unless specifically instructed.
- Expected increased urination with torsemide.
- Daily-weight technique and symptom monitoring.
- Antibiotic schedule and completion.
- Skin/leg monitoring, including worsening redness, swelling, drainage, or fever.
- Anticoagulation precautions.
- Fall precautions if the patient has experienced dizziness or blood-pressure changes.
- Follow-up appointments and laboratory monitoring.
- Teach-back of the medication changes before leaving the hospital.

Example 4 — COPD Exacerbation With Pneumonia and Acute on Chronic Heart Failure

Patient: 69-year-old man
Admission diagnosis: Acute hypoxic respiratory failure due to COPD exacerbation,

pneumonia, and volume overload

Discharge diagnoses:

- COPD exacerbation
- Community-acquired pneumonia
- Acute on chronic systolic heart failure
- Acute hypoxic respiratory failure, resolved
- Hypertension
- Tobacco use disorder

Hospital Course

The patient presented with increasing cough, wheezing, purulent sputum, and progressive dyspnea. He required supplemental oxygen on admission. Chest imaging demonstrated a left lower lobe infiltrate and pulmonary vascular congestion.

He was treated with systemic corticosteroids, bronchodilator therapy, antibiotics, and IV diuresis. Respiratory status gradually improved, and he was transitioned back to his baseline respiratory regimen. He was able to maintain appropriate oxygenation on his prescribed home oxygen requirement prior to discharge.

Heart-failure medications were adjusted because of borderline blood pressure during hospitalization. Diuretic dosing was increased temporarily, with improvement in volume status.

Discharge Medications

New medications

- **Prednisone 40 mg by mouth daily for 2 additional days** to complete COPD exacerbation treatment.
- **Doxycycline 100 mg by mouth twice daily for 3 additional days** to complete antibiotic treatment.
- **Furosemide 40 mg by mouth once daily**, increased from prior dose for volume management.

Continue

- Maintenance inhaler as reconciled.
- Rescue albuterol inhaler as needed.
- Carvedilol at the discharge dose.
- Other chronic medications as listed on the final medication reconciliation.

Stopped

- Previous lower-dose furosemide regimen, replaced by the new dose.
- Any duplicate inhaler therapy identified during medication reconciliation.

Discharge Instructions

- Complete prednisone and doxycycline courses.
- Use maintenance inhaler every day as prescribed; rescue inhaler is for breakthrough symptoms.
- Use home oxygen at the prescribed flow rate and during the prescribed activities/sleep periods.
- Continue heart-failure monitoring, including daily weights.
- Follow a low-sodium diet.
- Strongly recommend smoking cessation.
- Follow up with primary care, pulmonology, and cardiology as arranged.

Return Precautions

Seek emergency care for:

- Severe shortness of breath not relieved by prescribed rescue treatment.
- New or worsening chest pain.
- Blue/gray lips or severe difficulty speaking because of breathlessness.
- New confusion or inability to stay awake.
- Significant increase in oxygen requirement.
- High fever or rapidly worsening productive cough.
- Rapid weight gain with worsening edema or orthopnea.

Nursing Discharge Teaching / What Nursing Typically Relays

Nursing would generally reinforce:

- Correct use of maintenance versus rescue inhalers.
- Completion of antibiotics and steroids.
- Oxygen safety and prescribed use.
- Smoking-cessation counseling/resources.
- Daily weights and fluid/low-sodium instructions.
- Recognition of worsening respiratory symptoms.
- Correct furosemide dose and expected urinary frequency.
- Follow-up appointments.

- Confirmation that the patient has oxygen, inhalers, medications, and appropriate transportation/support at home.

Example 5 — Infective Endocarditis With Heart Failure and Prolonged IV Antibiotic Therapy

Patient: 58-year-old woman
Admission diagnosis: Native mitral-valve infective endocarditis complicated by acute heart failure
Discharge diagnoses:

- Mitral-valve infective endocarditis
- Acute on chronic heart failure with reduced EF
- Pulmonary edema, resolved
- Chronic kidney disease
- Anemia of chronic disease
- History of intravenous drug use, in remission

Hospital Course

The patient was admitted with fever, malaise, progressive dyspnea, and lower-extremity edema. Blood cultures repeatedly grew methicillin-susceptible *Staphylococcus aureus*. Echocardiography demonstrated a mitral-valve vegetation with significant mitral regurgitation.

She was initially treated with empiric IV antibiotics and subsequently narrowed to a pathogen-directed IV antibiotic regimen after infectious-disease consultation. Cardiothoracic surgery evaluated the patient and recommended continued medical therapy with close outpatient follow-up at this time.

Her hospitalization was complicated by acute pulmonary edema and decompensated systolic heart failure. She required aggressive IV diuresis with subsequent clinical improvement. Guideline-directed heart-failure therapy was cautiously restarted as blood pressure and renal function permitted.

At discharge, she was afebrile with cleared follow-up blood cultures, improved respiratory status, and stable renal function. She requires completion of a prolonged course of IV antimicrobial therapy through a supervised outpatient program.

Discharge Medications

New medications

- **Cefazolin IV at the infectious-disease-directed dose and schedule**, to complete the planned endocarditis treatment course.
- **Furosemide 40 mg by mouth once daily** for ongoing volume management.
- **Losartan 25 mg once daily**, initiated at a low dose because of improving blood pressure and renal function.

Continue

- Metoprolol succinate at the discharge dose.
- Other chronic medications after final medication reconciliation.

Stopped

- Empiric inpatient antibiotics that were no longer indicated after culture-directed therapy.
- NSAIDs — avoid unless specifically approved.

Discharge Instructions

- Complete the entire IV antibiotic course.
- Attend all infectious-disease and cardiology follow-up appointments.
- Obtain laboratory monitoring according to the outpatient antimicrobial-treatment plan.
- Keep vascular-access/PICC care appointments and follow instructions for the line.
- Do not independently stop or change IV antibiotics.
- Monitor daily weight and symptoms of heart failure.
- Follow a low-sodium diet and prescribed fluid plan.
- Follow any activity restrictions or dental/invasive-procedure recommendations provided by the infectious-disease/cardiology teams.

Return Precautions

Seek urgent medical attention for:

- Fever, chills, or recurrent systemic illness.
- New or worsening shortness of breath.
- Chest pain.
- Fainting or severe dizziness.
- New neurologic symptoms such as weakness, facial droop, difficulty speaking, or confusion.
- Rapidly worsening edema or weight gain.
- Bleeding or significant drainage/redness around the vascular-access site.
- Sudden severe back, chest, or abdominal pain.
- Any concern for a complication related to the IV line.

Nursing Discharge Teaching / What Nursing Typically Relays

Nursing typically reinforces:

- The **exact antibiotic schedule and duration**, while emphasizing that the full course must be completed.
- PICC/vascular-access care and infection-prevention practices.
- Which symptoms could indicate recurrent infection or heart failure.
- Daily weights and when to contact the clinical team.
- Medication reconciliation, particularly new heart-failure medications and discontinued inpatient antibiotics.
- Laboratory and infusion appointments.
- Who to contact after discharge for medication/line problems.
- Emergency symptoms requiring immediate evaluation.
- Teach-back: patient demonstrates understanding of antibiotic administration/line-care instructions or confirms the responsible caregiver can perform them correctly.
- Confirmation that home services, infusion supplies, transportation, and follow-up appointments are arranged before discharge.

Common Nursing Discharge-Communication Elements

Across complex medical discharges, nurses are commonly expected to **reinforce—not independently modify—the physician's discharge plan**. Typical elements include:

Area	What nursing commonly reinforces

Medication reconciliation	New medications, discontinued medications, dose/frequency changes, timing, and major precautions
Antibiotics	Indication explained in general terms, schedule, duration, importance of completing therapy, major warning symptoms
Heart failure	Daily weights, low-sodium diet, fluid instructions if ordered, edema/breathing changes, when to call
Diuretics	Expected increased urination, timing, dizziness/dehydration precautions, laboratory follow-up
Anticoagulants	Adherence, bleeding precautions, avoiding unapproved medications such as NSAIDs
Insulin/diabetes	Glucose monitoring, medication changes, hypoglycemia precautions
Respiratory disease	Inhaler technique, oxygen instructions, rescue-treatment plan, worsening respiratory symptoms
Lines/PICC	Site care, infection precautions, infusion schedule, what to do if the line malfunctions
Follow-up	Appointment dates, laboratory testing, specialty follow-up, transportation/access issues
Return precautions	Specific symptoms requiring a call versus urgent/emergency evaluation
Teach-back	Patient/caregiver explains the plan in their own words to verify understanding
Practical barriers	Ability to obtain medications, equipment, home services, transportation, and caregiver support

Important distinction: nurses generally should not create new medication instructions, change doses, interpret abnormal results independently, or contradict the prescribing clinician's discharge plan. When the patient has questions or the medication list appears inconsistent, nursing typically contacts the responsible prescribing team/pharmacist for clarification before discharge.